MILITARY SPECIFICATION SHEET

ELECTRON TUBES, CATHODE RAY
TYPES 5CP1A, 5CP1B, AND 5CP12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tubes described herein shall consist of this specification sheet and the latest issue of MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus with post accelerator

See figure 1
Mounting postion: Any

ABSOLUTE RATINGS:

Parameter:	Type phosphor	Ef	Ec1	ed	Eb1	Eb2	Eb3	Light output	Rg	Zd	Ehk	Eb3/Eb2	Barometric pressure reduced
Unit:		V	V dc	v	V dc	V dc	V dc	ftL(cd/m^2)	Meg	Meg	V dc	Ratio	mmHg
Maximum:	All	6.9	0	550	1,100	2,200	4,400	---	1.5	1.0	-125	2:3	87
Minimum:	1A, 1B	5.7	-200	---	---	1,500	1,500	15(51.4)	---	---	---	---	---
	12	---	-200	---	---	1,500	1,500	---	---	---	---	---	---
TEST CONDITIONS:	1A, 1B	6.3	Adjust	---	Focus	1,500	3,000	---	---	---	---	---	---
	12 (see note 7)	6.3	Adjust	---	Focus	2,000	4,000	---	---	---	---	---	---

GENERAL:

Qualification - Required (see note 1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
1141	Pressure (implosion)	All		---	---	---	---	---	---
1002	Barometric pressure, reduced	All	See note 3	---	---	---	---	---	---
5111	Vibration	All		---	---	Width	---	1	mm
1331	Direct-interelectrode capacitance	All	G1 to all	---	---	Cg1	---	12	pF
		All	k to all	---	---	Ck	---	13.5	pF
		All	D1 to D2	---	---	C1D2	---	3.5	pF
		All	D3 to D4	---	---	C3D4	---	3.5	pF
		All	D1 to all	---	---	CD1	---	15	pF
		1A, 12	D3 to all	---	---	CD3	---	15	pF
		1B	D3 to all	---	---	CD3	---	13	pF
		All	D1 to all except D2	---	---	C	---	13	pF
		All	D2 to all except D1	---	---	C	---	13	pF
		All	D3 to all except D4	---	---	C	---	13	pF
		All	D4 to all except D3	---	---	C	---	13	pF
5101	Neck and bulb alinement (electrostatic types)	All		---	---	dia	---	2.25	in.
5216	Cathode illumination	All		---	---	---	---	---	---
5246	Focusing voltage (zero bias)	1A, 1B,		---	---	Eb1	300	515	V dc
		12		---	---	Eb1	374	690	V dc
5248	Deflection factor	1A, 1B,	1D2; Eb3 = Eb2 = 1,500 V dc	---	---	DF	47	63	V dc/in.
		12	1D2; Eb3 = Eb2 = 2,000 V dc	---	---	DF	58	88	V dc/in.
5248	Deflection factor	1A, 1B,	3D4; Eb3 = Eb2 = 1,500 V dc	---	---	DF	41	55	V dc/in.
		12	3D4; Eb3 = Eb2 = 2,000 V dc	---	---	DF	51	77	V dc/in.
5248	Deflection-factor uniformity	All		---	---	---	---	---	---
	Quality conformance inspection, part 1								
5201	Electrode current (cathode)	1A, 1B	Light = 15 ftL (51.4 cd/m^2); Ib3 = 50 µA dc	0.65	II	Ik	---	1,000	µA dc
		12	Ib3 = 60 µA dc (see note 4)	0.65	II	Ik	---	1,000	µA dc
5201	Voltage breakdown	All		0.65	II	---	---	---	---
5201	Voltage breakdown (electrostatic types)	All		0.65	II	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 - Continued								
5206	Gas "cross" (electrostatic deflection)	1A, 1B	Light = 15 ftL ($51.4\ cd/m^2$), (see note 2)	0.65	II	---	---	---	---
		12	Ib3 = 60 µA dc (see note 2)	0.65	II	---	---	---	---
5106	Bulb, screen and faceplate blemishes	All		0.65	II	---	---	---	---
5221	Light output	1A, 1B	See note 2	0.65	II	Light	15 (51.4)	---	ftL (cd/m^2)
5223	Modulation	1A, 1B	Light = 15 ftL ($51.4\ cd/m^2$)	0.65	II	ΔEc	---	45	V dc
		12	Ib3 = 60 µA dc (see note 4)	0.65	II	ΔEc	---	55	V dc
5231	Spot position (electrostatic deflection)	All		0.65	II	---	---	25	mm
5231	Spot displacement (leakage)	All		0.65	II	displ	---	10	mm
5241	Grid-cutoff voltage	1A, 1B		0.65	II	Ec1	-22.5	-71	V dc
		12		0.65	II	Ec1	-30	-95	V dc
5251	Grid No. 1 leakage current	All		0.65	II	---	---	---	---
5251	Anode No. 2 leakage current	All		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1101	Secureness of base, cap, or insert	All		---	---	---	---	---	---
1301	Heater current	All		---	---	If	540	660	mA
5201	Electrode current (anode No. 1)	1A, 1B	Ec1 = 0	---	---	Ib1	-50	10	µA dc
		12	Ib3 = 60 µA dc (see note 4)	---	---	Ib1	-50	10	µA dc
5101	Base alinement (electrostatic types)	All	1D2; pin No. 5	---	---	---	---	---	---
5101	Side terminal alinement (electrostatic types)	All	1D2	---	---	---	---	---	---
5101	Neck and base alinement (electrostatic types)	All		---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
5101	Angle between traces	1A, 12		---	---	---	---	---	---
		1B		---	---	---	89.5	90.5	degrees
5216	Stray light emission (conventional types)	All	Eb2 = 2,200 V dc Eb3 = 4,400 V dc Ec1 = cutoff	---	---	---	---	---	---
5221	Screen	12		---	---	cbf D0.1(E) -D0.3(E)	550 80	--- 100	cb cb
5226	Line width A (electrostatic deflection	1A, 1B	Light = 15 ftL ($51.4\ cd/m^2$) (see note 5)	---	---	Width	---	0.75	mm
		12	Ib3 = 5 µA dc (see notes 4 and 6)	---	---	Width	---	0.40	mm
5226	Line width B (electrostatic deflection	1A, 1B	Light = 15 ftL ($51.4\ cd/m^2$) (see note 6)	---	---	Width	---	0.85	mm
		12	Ib3 = 5 µA dc (see notes 4 and 7)	---	---	Width	---	0.45	mm
5246	Focusing voltage at cutoff	1A, 1B		---	---	Eb1	345	515	V dc
		12		---	---	Eb1	460	690	V dc
5248	Deflection factor	1A, 1B	1D2	---	---	DF	58	80	V dc/in.
		12	1D2	---	---	DF	77	106	V dc/in.
5248	Deflection factor	1A, 1B	3D4	---	---	DF	50	68	V dc/in.
		12	3D4	---	---	DF	66	90	V dc/in.
5251	Heater-cathode leakage current	All		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test provisions	All	Group C; Eb2 = 2,200 V dc t = 500 hours	---	---	---	---	---	---
		1A, 1B	Light = 15 ftL ($51.5\ cd/m^2$); Eb3 = 4,400 V dc	---	---	---	---	---	---
		12	Eb3 = 4,400 V dc Ib3 = 5 µA dc	---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Life test end points	1A,	Line width A	---	---	Width	---	0.75	mm
		1B	Line width B			Width	---	0.85	mm
			Modulation	---	---	ΔEc	---	45	V dc
		12	Ib3 = 60 μA dc; modulation	---	---	ΔEc	---	55	V dc
			Ib3 = 5 μA dc; Line width A	---	---	Width	---	0.40	mm
			Ib3 = 5 μA dc; Line width B	---	---	Width	---	0.45	mm

NOTES:

1. The construction of this gun shall be the zero Ib1 type. The following information and materials are to be available with the four regular samples when application for qualification approval is made:

   (a) The electron gun drawing with significant dimensions.
   (b) A sample of the electron gun to be used in manufacture of tubes.

2. This test to be performed at the conclusion of the holding period.

3. The test is made with maximum voltage (Eg1 maximum negative voltage) applied to the base pins, and deflection electrodes if applicable, only; and pressure of 37 mmHg. Connections should be made in a manner that does not degrade the tubes electrical voltage breakdown characteristics. Satisfactory operation is the absence of arc-over and corona.

4. P12 screen. The tube can be severely and permanently damaged if the current density on the screen is allowed to rise too high in static tests of this sort. For this reason the length of time during which the screen is bombarded should be kept as short as possible compatible with the readings of specified current limits.

5. The deflecting plates shall be returned to anode No. 2 through a minimum of 2.5-megohm resistors. The light output shall be set at 15-ft lamberts (51.4 cd/m^2). The high-frequency scanning shall be applied to the deflecting plates nearest to the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low-frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter, in the direction perpendicular to the direction of high-frequency scanning. Readjustments may be made for best overall focus. The tubes shall be observed for deflection focusing, astigmatism, or spot ellipticity observable to the eye as evidenced by fuzziness due to the lack of sharpness of trace (usually around edges), bow-tying (irregular width of any single line when observed at different points), bowing of trace, other than that caused by curvature of bulb. Such defects shall not be excessive as determined by the government representative. This test for focus is to be made in addition to the line width measurements.

6. The same conditions shall be set up as described in note 5 except that the connection of deflecting elements to the low- and high-frequency scanning supplies shall be interchanged and the amplitudes adjusted to 90 percent of the maximum tube diameter in both directions without any adjustments of focus. An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 6 and the defocusing, astigmatism, etc., shall not be excessive as determined by a government representative.

7. P12 screen. All tests except screen persistence and flourescent brightness can be made at 50 percent duty cycle pulsing.

8. Revision letters are not used to denote changes due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Air Force - 99
DLA - ES

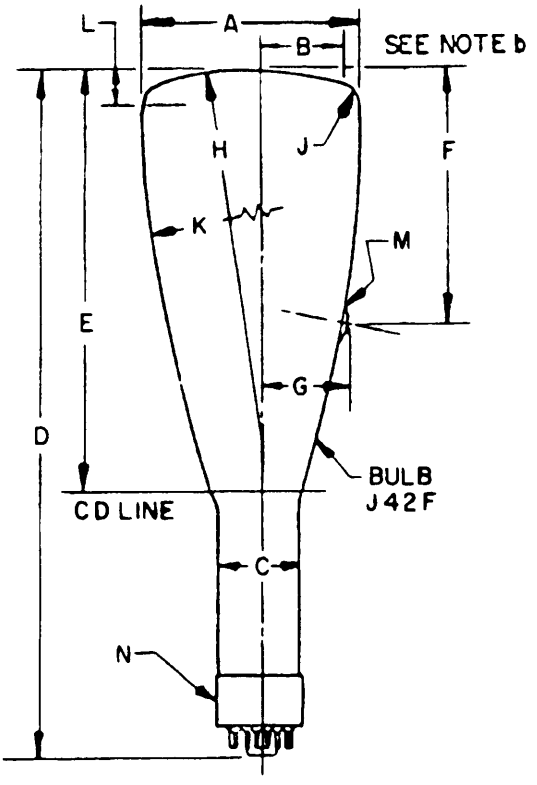
User activities:
Navy - AS, OS, MC, CG
Air Force - 11, 19

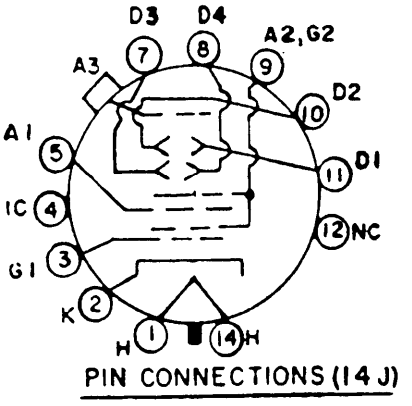
Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5960-3332)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.





PIN CONNECTIONS (14 J)

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.156 (130.96)	5.344 (135.74)
B	2.250 (57.15)	
C	1.937 (49.20)	2.063 (52.40)
D	16.375 (415.93)	17.125 (434.98)
E		10.375 (263.80)
F	6.000 (152.40)	6.500 (165.10)
Quality conformance inspection, part 3 (periodic check)		
M	Bulb contact: J1-22 (EIA)	
N	Base: B12-37 (EIA)	
Reference dimensions (see note a)		
G	2.313 (58.75)	
H	8.00 (203.20)	
J	.50 (12.70)	
K	27.813 (706.45)	
L	.820 (20.83)	

NOTES:
a. Reference dimensions are for information only and are not required for inspection purposes.
b. Useful screen radius.

FIGURE 1. Outline drawing of electron tube types 5CP1A, 5CP1B, and 5CP12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Naval Electronic Systems Command
Washington, DC 20363

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 2026l0-15T13:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-1/273H	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.